

Capturing the 3D Scan

Photon Robot

Summary

- Amazon.ca Listing: The Revopoint POP 3 Handheld 3D Scanner with Dual Axis Turntable - Advanced is available from the Revopoint Official Store on Amazon.ca, with limited stock (4 units as of last data). Search for the exact title, verify the turntable inclusion, and check for discounts (e.g., K4MNW7YR).
- Scanning Process: Use the POP 3 with the Dual Axis Turntable in Revo Scan 5 on your Mac to capture the Photon robot's 3D scan as a point cloud/mesh, stored locally. Apply scanning spray and markers for best results on shiny surfaces.
- Fusion 360 Import: Import the STL/PLY/OBJ file into Fusion 360, convert to a solid model using Mesh to BRep or manual sketching, and refine for reverse engineering. Use Meshmixer for mesh cleanup if needed.

Capture process details

To capture a 3D scan of the Photon educational robot using the Revopoint POP 3 with Dual Axis Turntable, follow these steps. The “image” (3D scan data) is captured as a point cloud or mesh in the Revo Scan 5 software on your Mac or a connected device.

Setup and Preparation

1. Environment:

- Location: Set up in a well-lit indoor area with diffuse lighting to avoid shadows. Avoid direct sunlight or harsh lighting, as it can interfere with the scanner's infrared and RGB cameras.
- Surface: Place the Dual Axis Turntable on a flat, stable surface. The turntable's 200 mm diameter and 5 kg load capacity easily accommodate the Photon robot's size (17.2 x 17 x 9.4 cm, 0.7 kg).
- Object Preparation:
 - Clean the Photon robot to remove dust or smudges.
 - If the robot's plastic surface is shiny, apply 3D scanning spray (e.g., Revopoint or AESUB sublimating spray) to create a matte finish, improving scan accuracy.

- Optionally, place 3D scanning markers on featureless areas (e.g., smooth panels) to aid tracking, as the POP 3 supports marker mode.

2. Hardware Setup:

- Connect the POP 3 scanner to your Mac via the USB Type-C cable (included in the kit). The scanner is macOS-compatible.
- Plug the Dual Axis Turntable into a power source (USB or AC adapter, if included) and connect it to the Revo Scan software via Bluetooth or USB for control.
- Place the Photon robot centred on the turntable. The turntable's 200 mm diameter provides ample space, and its $\pm 30^\circ$ tilt can help capture undercuts or angled surfaces.
- Position the scanner 15–40 cm from the robot, as recommended for optimal accuracy (0.05 mm precision).

3. Software Setup:

- Download and install Revo Scan 5 from the Revopoint website (www.revopoint3d.com) on your Mac. Ensure your Mac meets the software's system requirements (macOS 10.15 or later, 8GB RAM minimum).
- Open Revo Scan 5 and calibrate the scanner using the included calibration board and on-screen instructions. Calibration ensures precision down to 0.05 mm.
- Select Fixed Scan Mode with the turntable for automated scanning of the Photon robot. Alternatively, use Handheld Mode for manual scanning of complex areas, but the turntable is recommended for consistency.

Lighting Setup Instructions

1. Position the Turntable:

- Place the Revopoint Dual Axis Turntable on a flat, stable surface in your room (e.g., a desk or table).
- Center the Photon robot on the turntable, ensuring it's secure (use non-slip pads if needed, as the robot weighs ~0.7 kg).

2. Arrange the Lights:

- Position the two softbox or ring lights on their stands at 45-degree angles to the turntable, approximately 50–70 cm from the Photon robot.
 - Height: Set lights at 1–1.5 m, slightly above the robot, to illuminate the top and sides evenly.
 - Angle: Angle lights downward to cover the robot's 17.2 x 17 x 9.4 cm volume, avoiding direct projection into the POP 3 scanner's lenses.

- **Brightness:** Set to 50–70% brightness initially to avoid overexposure. Adjust based on Revo Scan 5's live preview (if the image looks washed out, lower brightness; if too dark, increase).
 - **Diffusion:** Attach white diffusers (included with Neewer lights) to soften light. If using diffusion fabric, drape it loosely around the setup, leaving an opening for the scanner (like a partial light tent).
3. Minimize Ambient Light:
- Close curtains or blinds to block external light (e.g., sunlight or streetlights), as the POP 3 performs best in controlled conditions.
 - Turn off room lights or dim them to avoid conflicting light sources. The softboxes/ring lights should provide all illumination.
4. Prepare the Photon Robot:
- Apply 3D scanning spray (e.g., Revopoint or AESUB) to the robot's shiny plastic surface to reduce reflections, which are exacerbated in uneven lighting. Spray lightly from 20–30 cm away for a thin, even coat.
 - Place 3.0mm scanning markers on featureless areas (e.g., smooth panels) to aid tracking, especially in low-light areas of your room.
5. Scanner Setup:
- Connect the POP 3 scanner to your Mac via USB-C and position it 15–40 cm from the robot, as recommended for 0.05 mm precision.
 - Open Revo Scan 5 on your Mac (macOS 10.15 or later), calibrate the scanner using the included calibration board, and select Fixed Scan Mode with the turntable.
 - Set the turntable speed to 30 seconds/rotation for detailed scans and enable Color Scan to capture texture.
6. Test and Adjust:
- Start a test scan in Revo Scan 5 and monitor the live preview. If the point cloud has gaps or noise:
 - **Adjust Light Position:** Move lights closer or farther (within 50–100 cm) to balance illumination.
 - **Tweak Brightness:** Increase/decrease brightness to optimize contrast in the preview.
 - **Check Spray/Markers:** Ensure the robot's surface is matte and markers are visible.
 - Perform 2–3 scans with different turntable tilts ($\pm 30^\circ$) to capture all surfaces (e.g., top, sides, undercuts).

Scanning Process

1. Configure Scan Settings:

- In Revo Scan 5, choose High Precision Mode for the best detail (0.05 mm accuracy) and enable Color Scan to capture the robot's texture (useful for identifying components like buttons or sensors).
 - Set the turntable speed to 18–30 seconds per rotation for detailed scans (slower speeds improve accuracy).
 - If using markers, enable Marker Mode in the software to track them for better alignment.
2. Capture the Scan:
- Start the scan in Revo Scan 5 using the scanner's touch-sensitive buttons or software controls.
 - The turntable will rotate the Photon robot, and the POP 3 will capture a point cloud (a collection of 3D data points representing the object's surface). The software automatically stitches frames using the 9-axis IMU for stable tracking.
 - Perform multiple scans (2–3 orientations) by tilting the turntable ($\pm 30^\circ$) or repositioning the robot to capture undercuts, wheels, or ports. Revo Scan 5 auto-aligns scans from different angles.
 - Monitor the live preview in Revo Scan 5 to ensure all areas are captured. If gaps appear, adjust the scanner distance or apply more scanning spray/markers.
3. Post-Processing in Revo Scan 5:
- After scanning, use Revo Scan 5's one-click model processing to merge scans, remove noise, and fill holes. The software's intuitive interface simplifies this for beginners.
 - Smooth the mesh to reduce artifacts, but avoid over-smoothing to preserve details like the robot's edges or buttons.
 - Export the final model as an STL, PLY, or OBJ file, all of which are compatible with Fusion 360.

Where the Image is Captured

- The “image” (3D scan) is captured as a point cloud or mesh within the Revo Scan 5 software on your Mac during the scanning process. The data is stored locally on your Mac's hard drive, typically in a project folder created by Revo Scan 5 (default location depends on your macOS settings, e.g., Documents or Desktop).
- The point cloud is generated by the POP 3's structured light technology, which projects infrared patterns onto the Photon robot and records reflections with its depth and RGB cameras. The turntable ensures consistent rotation, reducing manual effort and improving stitch accuracy.
- After processing, the final mesh file (STL/PLY/OBJ) is saved to your chosen directory, ready for import into Fusion 360.

Clean up with Blender or MeshLab

If there are gaps or other issues with the mesh, follow this workflow to clean it up.

1. Import Scan:
 - In Blender: File > Import > STL/PLY/OBJ, select the POP 3 scan file from Revo Scan 5.
 - In MeshLab: File > Import Mesh, select the file.
2. Clean Up:
 - Blender: Use Mesh > Clean Up > Fill Holes and Modifiers > Decimate (set ratio to 0.5–0.7 to reduce polygons).
 - MeshLab: Apply Filters > Cleaning and Repairing > Close Holes and Simplify Mesh.
3. Smooth:
 - Blender: Enter Sculpt Mode and use the Smooth Brush.
 - MeshLab: Use Filters > Smoothing > Laplacian Smooth.
4. Export:
 - Export as STL (preferred for Fusion 360) to your Mac's Documents folder.

Importing the Scan into Autodesk Fusion 360 on macOS

To import the 3D scan of the Photon robot into Autodesk Fusion 360 on your Mac and begin reverse engineering, follow these steps:

Prerequisites

- Fusion 360 Installation: Ensure Fusion 360 is installed on your Mac (macOS 10.15 or later). Students and educators can access a free license through Autodesk's Education Community (www.autodesk.com/education) (www.autodesk.com/education). Otherwise, a subscription costs ~\$500/year.
- System Requirements: Fusion 360 requires at least 4GB RAM (8GB recommended) and a compatible GPU. Verify your Mac meets these specs for smooth performance with 3D scan data.
- Scan File: You should have the processed scan from Revo Scan 5 saved as an STL, PLY, or OBJ file (STL is recommended for CAD workflows).

Import Process

1. Open Fusion 360:

- Launch Fusion 360 on your Mac and create a new project or open an existing one.
 - Ensure you're in the Design workspace (default for modeling).
2. Import the Scan File:
- Go to Insert > Insert Mesh in the top menu.
 - Browse to the location of your STL/PLY/OBJ file (e.g., in your Documents folder where Revo Scan 5 saved it).
 - Select the file and click Open. Fusion 360 will load the mesh, which may take a few seconds depending on file size (the Photon robot's scan should be manageable, as it's a small object,).
 - Adjust the Units if prompted (set to mm to match the Photon robot's dimensions: 172 x 170 x 94 mm).
3. Position and Scale the Mesh:
- The imported mesh appears in the Fusion 360 canvas. Use the Move/Copy tool (under Modify) to position the mesh at the origin or align it as needed.
 - Check the mesh's scale by measuring a known dimension (e.g., the robot's 172 mm length) using the Inspect > Measure tool. If scaling is off, use Modify > Scale to adjust the mesh to match real-world dimensions.
4. Convert Mesh to Solid Model for Reverse Engineering:
- Fusion 360 works best with solid bodies for CAD modelling, but the imported scan is a mesh (a collection of triangles). To convert it:
 - Switch to the Mesh workspace (top menu dropdown).
 - Use Mesh > Reduce to simplify the mesh if it's too dense (high polygon counts can slow down Fusion 360). Aim for a balance between detail and performance (e.g., reduce by 20–50% for the Photon robot's simple geometry).
 - Go to Create > Mesh to BRep (Body Representation) to attempt converting the mesh to a solid body. Note: This feature may struggle with complex or noisy meshes, especially if the scan has holes or artifacts.
 - Alternative Workflow for Complex Meshes:
 - If Mesh to BRep fails, manually trace the mesh to create a solid model:
 - Create a Sketch on a plane (e.g., XY plane) and use the mesh as a reference to draw 2D profiles (e.g., for the robot's body or wheels).
 - Use tools like Extrude, Revolve, or Loft to build solid features based on the sketches.
 - For curved surfaces (e.g., the robot's rounded edges), use the Form workspace to sculpt T-Splines over the mesh, then convert to a BRep solid.
 - This manual process is common for reverse engineering, as consumer-grade scans like those from the POP 3 may require cleanup.
5. Refine the Model:

- Use Inspect > Measure to verify dimensions against the Photon robot's known specs (172 x 170 x 94 mm).
 - Add features like holes, ports, or buttons by creating sketches and using Extrude or Hole tools, referencing the mesh for placement.
 - Smooth surfaces or simplify geometry using Modify > Fillet or Simplify tools to prepare the model for 3D printing or further analysis.
6. Save and Export:
- Save your Fusion 360 project to the cloud (Autodesk 360) or locally as an .f3d file.
 - Export the final model as an STL for 3D printing or STEP/IGES for other CAD software, if needed, via File > Export.

Tips for Success in Fusion 360

- Clean the Mesh: If the scan has noise or holes, use Meshmixer (free, available for macOS) to repair the mesh before importing into Fusion 360. Import the STL/PLY/OBJ into Meshmixer, use Analysis > Inspector to fix holes, and smooth with Sculpt tools.
- Performance: The Photon robot's scan should be manageable, but dense meshes can slow down Fusion 360. Reduce polygon count in Revo Scan 5 or Meshmixer if needed.
- Learning Curve: Fusion 360 has a learning curve for reverse engineering. Watch Autodesk's Fusion 360 YouTube tutorials or Revopoint's scanning tutorials for guidance on mesh-to-CAD workflows ().
- Mac Compatibility: The POP 3 and Revo Scan 5 are fully compatible with macOS, and Fusion 360 runs smoothly on modern Macs (). Ensure your Mac has sufficient RAM (8GB minimum) to handle scan processing.

Best Practices for Scanning the Photon Robot

- Turntable Use: The Dual Axis Turntable is ideal for the Photon robot's size. Set it to a slow speed (e.g., 30 seconds/rotation) and tilt it to capture undercuts like the robot's wheels or bottom.
- Shiny Surfaces: The robot's plastic surface may reflect the scanner's infrared light, causing data loss. Apply scanning spray evenly (avoid over-application) or use markers for better tracking.
- Multiple Scans: Scan the robot in 2–3 orientations (e.g., upright, tilted forward, tilted sideways) to capture all surfaces. Revo Scan 5's auto-alignment simplifies merging these scans.
- Calibration: Recalibrate the scanner if accuracy drops or if you move to a new environment. Use the included calibration board and follow Revo Scan 5's prompts.

- **Software Updates:** Check for Revo Scan 5 updates on Revopoint's website, as users report improved performance with newer versions (e.g., v5.02.).

Troubleshooting

- **Missing Components:** Some Amazon.ca buyers reported missing tripods or phone holders. Inspect the package upon delivery and contact Revopoint's support (support@revopoint3d.com) or Amazon if items are missing. Revopoint is noted for responsive support.
- **Shiny Surface Issues:** If scans are incomplete, apply more scanning spray or add markers. Avoid scanning in bright sunlight.
- **Fusion 360 Mesh Issues:** If the mesh is too noisy, use Meshmixer to clean it before importing. If Mesh to BRep fails, manual sketching is a reliable fallback.
- **Turntable Alignment:** Center the Photon robot on the turntable using the included markers or a custom 3D-printed adapter (available on Thingiverse for the Portable Turntable).